

Azure Fundamentals & Core Services

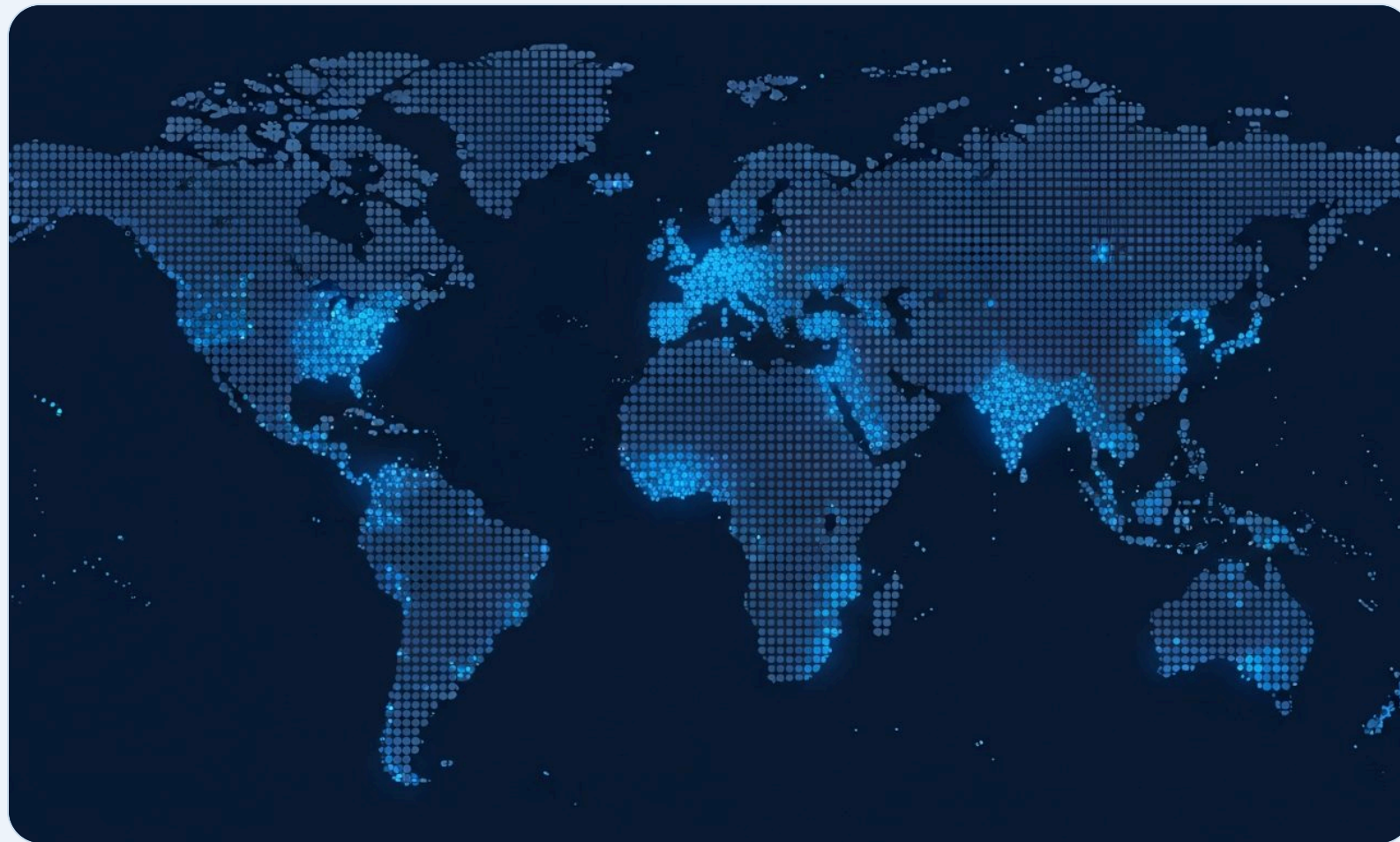
A hands-on workbook for engineers moving from Google Cloud to Microsoft Azure.

— **Lecture by Imran Sayyed**

The three major clouds, side by side

Capability	AWS	Azure	Google Cloud
Virtual machines	EC2	Virtual Machines	Compute Engine
Object storage	S3	Blob Storage	Cloud Storage
Serverless	Lambda	Functions	Cloud Functions
Identity	IAM	Entra ID + RBAC	Cloud IAM
Grouping unit	Account / Tags	Resource Group	Project

Azure global infrastructure



Geography

A market containing one or more regions, keeping data residency in-country.

Region

A set of datacenters in one area — the same idea as a GCP region.

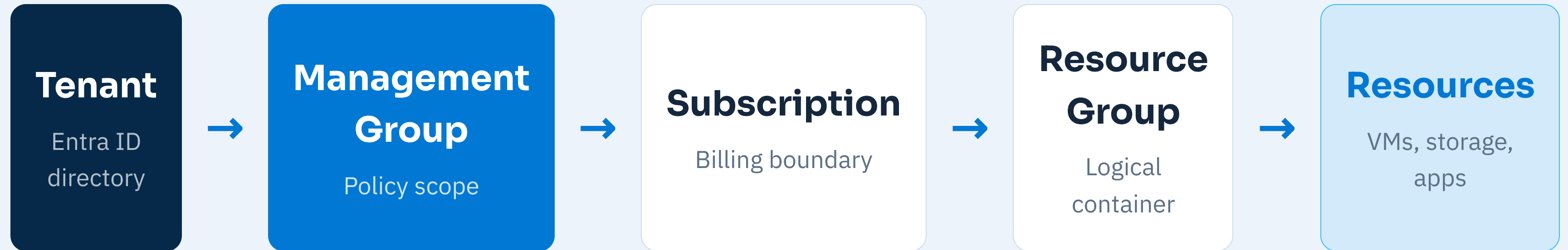
Availability Zone

Physically separate datacenters inside a region, like GCP zones.

Region Pair

Two regions paired for replication and disaster recovery.

How Azure organizes everything



Read it top-down: identity lives in the tenant, money is tracked per subscription, and resources are grouped for shared lifecycle.

What is a tenant?

A **tenant** is a dedicated instance of Microsoft Entra ID — your organization's identity boundary.

It holds every user, group, and app registration. One company normally has exactly one tenant.

All subscriptions trust this tenant for sign-in and authentication.

COMING FROM GCP

Tenant $\approx$ Organization

Like a GCP Organization node, the tenant is the top-level identity and trust root for everything below it.

What is a subscription?

A **subscription** is a billing and access boundary that holds your resource groups.

Every resource you deploy is charged to exactly one subscription, and quotas are enforced at this level.

Teams often split work into separate subscriptions for dev, test, and production.

COMING FROM GCP

Subscription $\approx$ Project (for billing)

It plays the billing and quota role of a GCP Project, but the grouping of resources happens one level down.

What is a resource group?

A **resource group** is a logical container for resources that share the same lifecycle — deploy, manage, and delete them together.

Same lifecycle

Delete the group and every resource inside it is removed.

One region tag

The group has a location for metadata; resources can live elsewhere.

Access & tags

Apply RBAC roles and cost tags at the group level.

Resource Group vs GCP Project

Question	Azure Resource Group	GCP Project
Billing?	No — billing is the subscription	Yes — billing lives here
Groups resources?	Yes — primary purpose	Yes
Lifecycle scope?	Delete group = delete contents	Delete project = delete contents

A GCP Project splits into two Azure ideas: the **subscription** for billing and the **resource group** for grouping.

Create your first resource group

- 1 Sign in at portal.azure.com.
- 2 Search **Resource groups**, then click + **Create**.
- 3 Pick a subscription and name it **rg-learn-01**.
- 4 Choose a region such as **East US**.
- 5 Click **Review + create**, then **Create**.

PORTAL PATH

Home → Resource groups → Create

EXPECTED RESULT

rg-learn-01 appears in your resource group list.

WATCH OUT

The name must be unique within the subscription.

Who you are, and what you can touch

Azure splits access into two questions: authentication handled by Entra ID, and authorization handled by RBAC role assignments.

Entra ID

Identity & sign-in

RBAC

Permissions on resources

Microsoft Entra ID

Entra ID (formerly Azure AD) is the cloud identity service that authenticates users and apps before they reach any resource.

Users & groups

People and teams that sign in to your organization.

App registrations

Service identities for apps and automation.

Conditional access

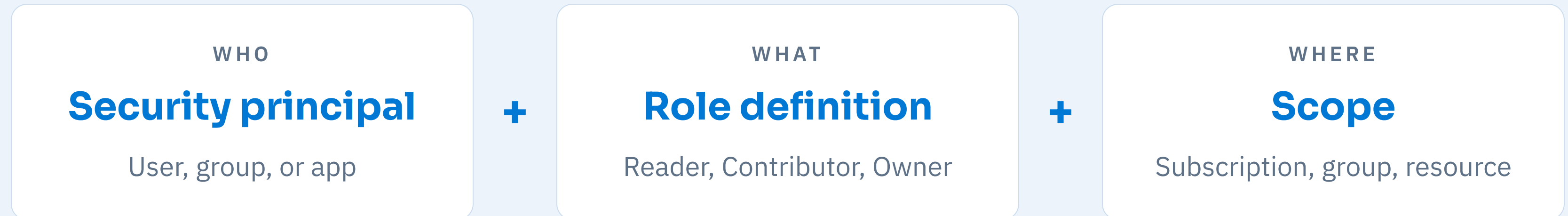
Policies like MFA that gate every sign-in.

GCP bridge: think Cloud Identity plus the directory behind Google Workspace.

How RBAC works

RBAC = Role-Based Access Control

A role assignment connects three things — answer all three and access is granted.



Scope is inherited downward, so a role at the subscription flows to every resource group beneath it.

Azure RBAC vs GCP IAM

Concept	Azure RBAC	GCP IAM
Identity	User / group / service principal	User / group / service account
Permission set	Role definition	Role
Common roles	Reader, Contributor, Owner	Viewer, Editor, Owner
Scope levels	Mgmt group → sub → RG → resource	Org → folder → project → resource

Assign the Reader role

- 1 Open **rg-learn-01** and select **Access control (IAM)**.
- 2 Click + **Add** → **Add role assignment**.
- 3 Choose the **Reader** role, then click **Next**.
- 4 Select a member, then **Review + assign**.

PORTAL PATH

Resource group → Access control (IAM) → Add

EXPECTED RESULT

The member can view but not change resources in the group.

WATCH OUT

Role changes can take a few minutes to take effect.

Renting servers in the cloud

Compute is where your code runs. The model is the same across clouds: choose a machine size, an operating system, and a network — then pay while it runs.

IaaS

You manage the OS

Pay-as-you-go

Billed per second running

Elastic

Resize or scale on demand

Azure Virtual Machines

A **Virtual Machine** is an on-demand, scalable server you control end to end — OS, runtime, and software.

- **Image:** the OS, such as Ubuntu or Windows Server.
- **Size:** the vCPU and memory tier, like B2s or D4s.
- **Disks:** managed OS and data disks.
- **Networking:** a virtual network, NIC, and public IP.

You patch and secure the OS yourself — that is the IaaS trade-off for full control.

VM vs Compute Engine vs EC2

Concept	Azure VM	Compute Engine	AWS EC2
Machine sizing	VM size (B2s)	Machine type	Instance type
OS template	Image	Image	AMI
Block storage	Managed Disk	Persistent Disk	EBS
Autoscale group	Scale Set	Managed Instance Group	Auto Scaling Group

Create a Linux virtual machine

- 1 Search **Virtual machines** → **+ Create**.
- 2 Use **rg-learn-01** and name it **vm-web-01**.
- 3 Pick **Ubuntu** and size **B1s**.
- 4 Set an **SSH key** for authentication.
- 5 Click **Review + create**.

PORTAL PATH

Home → Virtual machines → Create

EXPECTED RESULT

Deployment completes and the VM shows status Running.

WATCH OUT

Stop the VM when idle — it bills while running.

Reading the VM create screen

Create a virtual machine

Resource group	rg-learn-01
Image	Ubuntu 22.04 LTS
Size	Standard B1s
Authentication	SSH public key

Tabs to know

Basics, Disks, Networking, Management — most labs only need Basics.

Inbound ports

Allow SSH (22) to connect; open HTTP (80) only if hosting a site.

Validation

The portal runs checks before the Create button unlocks.

Connect to your VM over SSH

- 1 Open **vm-web-01** and copy its **public IP**.
- 2 Select **Connect** → **SSH** for the exact command.
- 3 From your terminal, run the SSH command with your key.
- 4 Accept the host fingerprint to land on the shell.

COMMAND

```
ssh -i key.pem azureuser@<public-ip>
```

EXPECTED RESULT

A Linux shell prompt for the azureuser account.

WATCH OUT

Connection refused usually means port 22 is not allowed.

Where your data lives

Cloud storage comes in shapes: objects for unstructured files, disks for VMs, and shares for file systems. We focus on object storage — the workhorse for apps.

Object

Blobs — files & media

Disk

Managed disks for VMs

File

SMB / NFS shares

The storage account

A **storage account** is the top-level namespace that holds all your storage services under one unique endpoint.



The account name forms your endpoint: name.blob.core.windows.net

Blob storage & access tiers

Blobs live inside containers and store any unstructured data — images, backups, logs, or video.

Hot

Frequent access, highest storage cost, lowest read cost.

Cool

Infrequent access, kept at least 30 days.

Archive

Cheapest storage, rehydrate before reading.

Same tiering idea as GCP Standard, Nearline, and Coldline classes.

Blob vs Cloud Storage vs S3

Concept	Azure Blob	Cloud Storage	AWS S3
Grouping unit	Container	Bucket	Bucket
Stored item	Blob	Object	Object
Hot tier	Hot	Standard	Standard
Archive tier	Archive	Coldline / Archive	Glacier

Create a storage account

- 1 Search **Storage accounts** → **+ Create**.
- 2 Use **rg-learn-01** and a globally unique name.
- 3 Pick a region and **Standard** performance.
- 4 Choose **LRS** redundancy, then create.

PORTAL PATH

Home → Storage accounts → Create

EXPECTED RESULT

A storage account ready with its blob endpoint.

WATCH OUT

The name must be lowercase, 3–24 chars, globally unique.

Create a blob container

- 1 Open your storage account and select **Containers**.
- 2 Click + **Container** and name it **uploads**.
- 3 Keep access level **Private**.
- 4 Click **Create**.

PORTAL PATH

Storage account → Containers → + Container

EXPECTED RESULT

An empty private container named uploads.

WATCH OUT

Public access is off by default — keep it that way for labs.

Upload and read a blob

- 1 Open the **uploads** container.
- 2 Click **Upload** and pick a small image.
- 3 Open the blob and copy its **URL**.
- 4 Generate a **SAS token** to access it temporarily.

PORTAL PATH

Container → Upload → Generate SAS

EXPECTED RESULT

The blob opens in a browser through its SAS URL.

WATCH OUT

The plain blob URL returns an error without a SAS on a private container.

Code without managing servers

Serverless flips the model: you ship a function, the platform runs it on demand, and you pay only for the execution time. No VMs to patch.

Event-driven

Triggers start the code

Scale to zero

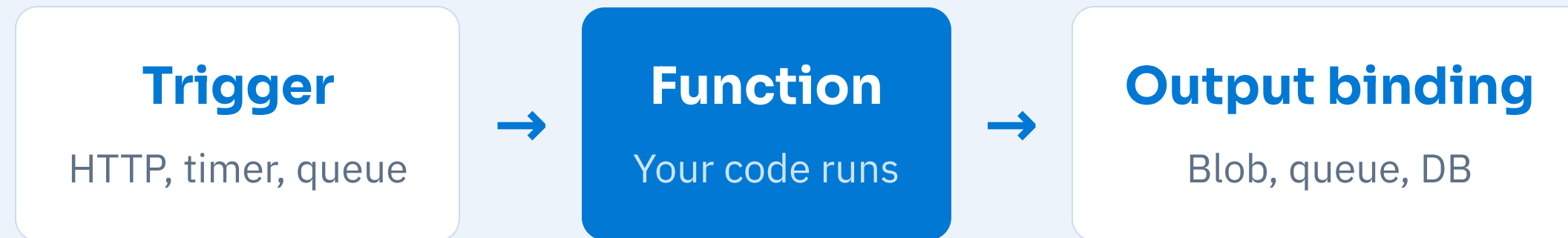
No cost when idle

Per-execution

Billed by run & memory

Azure Functions

An **Azure Function** is a small piece of code that runs in response to a trigger, hosted inside a Function App.



The Consumption plan scales to zero, so you pay nothing while no triggers fire.

Azure Functions vs Cloud Functions

Concept	Azure Functions	Cloud Functions
Host unit	Function App	Individual function
Free-scale plan	Consumption plan	Pay-per-invocation
Common trigger	HTTP trigger	HTTP trigger
Bindings	Input & output bindings	Manual SDK calls

Create an HTTP function

- 1 In the portal, search **Function App** and click **+ Create**.
- 2 Pick group **rg-learn-01**, runtime stack **Python**, and the **Consumption** plan.
- 3 Click **Review + create**, then wait for the deploy to finish.
- 4 Add a function using the **HTTP trigger** template.
- 5 Click **Get function URL** and open it in your browser.

PORTAL PATH

Function App → Functions → Create

EXPECTED RESULT

Opening the URL returns a hello response from your Python function.

GCP BRIDGE

Like a Python Cloud Function — same HTTP trigger, same handler idea.

Knowing what your systems are doing

Observability rests on three signals: metrics for numbers over time, logs for events, and alerts that tell you when something needs attention.

Metrics

CPU, memory, requests

Logs

Queried with KQL

Alerts

Act on thresholds

Azure Monitor

Azure Monitor is the unified platform that collects, analyzes, and acts on telemetry from every resource.

Metrics Explorer

Plot numeric series like CPU percentage in near real time.

Log Analytics

Query collected logs using the Kusto query language.

Alerts

Fire when a metric or log condition crosses a threshold.

Application Insights

Deep tracing for web apps and APIs.

Azure Monitor vs Cloud Operations

Capability	Azure Monitor	Google Cloud
Metrics	Metrics Explorer	Cloud Monitoring
Logs	Log Analytics (KQL)	Cloud Logging
App tracing	Application Insights	Cloud Trace
Query language	KQL	Logging query language

Watch VM metrics & set an alert

- 1 Open **vm-web-01** and select **Metrics**.
- 2 Pick the **Percentage CPU** metric.
- 3 Go to **Alerts → Create alert rule**.
- 4 Set CPU > 80% and add an email action group.

PORTAL PATH

VM → Monitoring → Metrics / Alerts

EXPECTED RESULT

A live CPU chart and an alert rule in Enabled state.

WATCH OUT

Metrics can lag a few minutes behind real activity.

Connecting and protecting resources

A virtual network is your private slice of the cloud. Subnets divide it, and security rules decide which traffic is allowed in and out.

VNet

Private address space

Subnet

Segments inside a VNet

NSG

Firewall rules

Network Security Groups

An **NSG** is a set of rules that allow or deny traffic to subnets and network interfaces — a built-in firewall.

Priority	Port	Protocol	Source	Action
100	22	TCP	My IP	Allow
200	80	TCP	Any	Allow
65500	Any	Any	Any	Deny

Lower priority numbers win, and a default deny sits at the bottom of every NSG.

NSG vs VPC Firewall Rules

Concept	Azure NSG	GCP VPC Firewall
Private network	Virtual Network (VNet)	VPC network
Attaches to	Subnet or NIC	Network (by tags)
Rule ordering	Priority number	Priority number
Default	Deny inbound	Deny inbound

Add an inbound NSG rule

- 1 Open the NSG attached to **vm-web-01**.
- 2 Select **Inbound security rules** → **+ Add**.
- 3 Allow port **80** (TCP) with priority **200**.
- 4 Save, then browse to the VM's public IP.

PORTAL PATH

NSG → Inbound security rules → Add

EXPECTED RESULT

HTTP traffic now reaches the VM on port 80.

WATCH OUT

A lower priority number overrides a higher one — order matters.

A typical Azure web app

RESOURCE GROUP: RG-APP-PROD

Users

HTTPS



VM + NSG

Web tier



Blob Storage

Assets



Functions

Background jobs

Azure Monitor watches every tier · Entra ID + RBAC control access

Check yourself

1. Which holds billing — subscription or resource group?

Answer: the subscription.

2. What three parts make an RBAC role assignment?

Answer: principal, role definition, and scope.

3. Which blob tier is cheapest to store but slow to read?

Answer: the Archive tier.

4. In an NSG, does a lower or higher priority number win?

Answer: the lower number wins.

Azure ↔ GCP cheat sheet

Area	Azure	GCP
Billing scope	Subscription	Project
Grouping	Resource group	Project
Identity	Entra ID	Cloud Identity
Compute	Virtual Machine	Compute Engine
Object storage	Blob Storage	Cloud Storage
Serverless	Azure Functions	Cloud Functions
Networking	VNet + NSG	VPC + Firewall

Build a mini web stack

Combine everything: provision a small end-to-end environment in a single resource group.

- 1 Create resource group **rg-challenge**.
- 2 Deploy a VM running a simple web server.
- 3 Open port 80 with an NSG rule.
- 4 Store an asset in a blob container.
- 5 Add a CPU alert in Azure Monitor.
- 6 Delete the group to clean up everything.

Common interview questions

Explain the Azure resource hierarchy from tenant to resource.

How does RBAC inheritance work across scopes?

When would you choose Functions over a VM?

Compare blob access tiers and their trade-offs.

How do NSG rules decide whether traffic is allowed?

What is the difference between metrics and logs?

Tip: anchor each answer to the GCP equivalent you already know.

▲ WRAPPING UP

You can now navigate Azure

Organize

Tenant → subscription → group

Secure

Entra ID & RBAC

Build

VMs, storage, functions

Operate

Monitor & network securely

— Lecture by Imran Sayyed